



Mira
The variable star Mira is circled in the center of this image. It is here at its brightest and visible to the naked eye. The two brightest “stars” in this view are, in fact, planets: Jupiter is at right, with Saturn above and to its left.

■ SOUTHERN LATITUDES

The constellation Cetus is overhead. Mira, one of its stars, is a well-known variable. It is a red giant whose brightness changes over about 11 months. It is about magnitude 3 at brightest and within naked-eye visibility. At its dimmest, about magnitude 10, it is visible only through a telescope. Mira is the prototype for long-period variables known as Mira variables.

Pisces and Aries are on either side of north. Pisces, depicting two fish joined with a cord, is a faint constellation. Its brightest star, alpha (α), marks the knot that ties the fish together; a telescope reveals it as two stars. Below Pisces and to the northeast are Pegasus and Andromeda, which are linked by the Great Square of Pegasus. The Andromeda Galaxy remains high enough to be seen.

November offers the chance to see Andromeda and Perseus together. Perseus, who killed the sea monster Cetus and

prevented it from devouring Andromeda, is near the horizon in the northeast. The star Beta (β) Persei, called Algol, is a well-known eclipsing binary. It consists of two stars orbiting around each other. Together they are magnitude 2.1, but when the fainter passes in front of its brighter companion, the brightness drops to magnitude 3.4. This change can be detected by the naked eye. The dip and return to full brightness takes about 10 hours every 69 hours.

Achernar in Eridanus is center stage to the south. To its west is Fomalhaut in Piscis Austrinus, the southern fish. The fish is the parent of the two in Pisces and is the recipient of water pouring from Aquarius’s jug at right. Four birds—Phoenix, Grus (the crane), Tucana (the toucan), and Pavo (the peacock)—are in the southwest. The Small Magellanic Cloud is seen in Tucana, and to its left is the Large Magellanic Cloud in Dorado. Both irregular galaxies are visible to the naked eye. Individual clusters and nebulae are seen with binoculars or a telescope. Canopus, the white supergiant and second-brightest star of all, is in the southeast. Beyond, in the east, is Sirius (Canis Major), the brightest of all. The presence of Taurus, Orion, and Canis Major above the horizon is a sign that summer is approaching.



Pisces
The distinguishing feature of Pisces is the ring of seven stars that marks the body of one of the two fish. Alpha (α) Piscium (lower left) marks the knot where the cords from the fish are tied together.

